$yS = (y \times 1)^{\#}(S)$

by

§1.616 — pre-composing a relation by a map is an inverse image

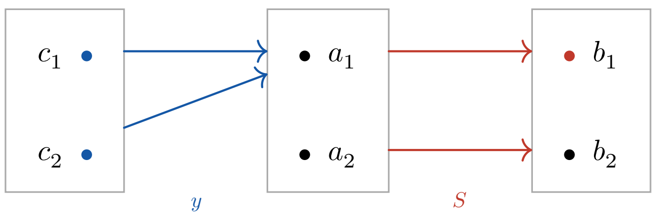
For a **map** $y : C \to A$ and a **relation** $S : A \to B$, the composite $yS : C \to B$, as a subobject of $C \times B$, is the inverse image of S along $y \times 1$:

$$yS \quad = \quad (y \times 1)^{\#}(S)$$

where $y \times 1 = y \times 1_B : C \times B \to A \times B$, $(c, b) \mapsto (y(c), b)$. As a pullback:

$$\begin{tikzcd}
C \times B \arrow[rr, "y \times 1"] & & A \times B \\
yS & & S \arrow[ll, "(y \times 1)^{\#}"'] \\
\text{subobject of } C \times B & & \text{subobject of } A \times B
\end{tikzcd}$$

§0.1. In Set



As a composite: $c\ (yS)\ b \iff (y(c), b) \in S$. Both $c_1, c_2 \mapsto a_1$ and $a_1\ S\ b_1$, so

$$yS = \{(c_1, b_1), (c_2, b_1)\} \quad (c_2 \to a_1 \to b_1).$$

As an inverse image: $y \times 1$ sends $(c, b) \mapsto (a_1, b)$, so the preimage of S is

$$(y \times 1)^{\#}(S) = \{(c, b) : (y(c), b) \in S\} = \{(c_1, b_1), (c_2, b_1)\}.$$

Same subset of $C \times B$.

> ⋆ **Punchline**
>
> Pre-composing S by the map y is pulling S back along $y \times 1$. So inverse images preserving unions gives, for free, $y(S \cup T) = yS \cup yT$.